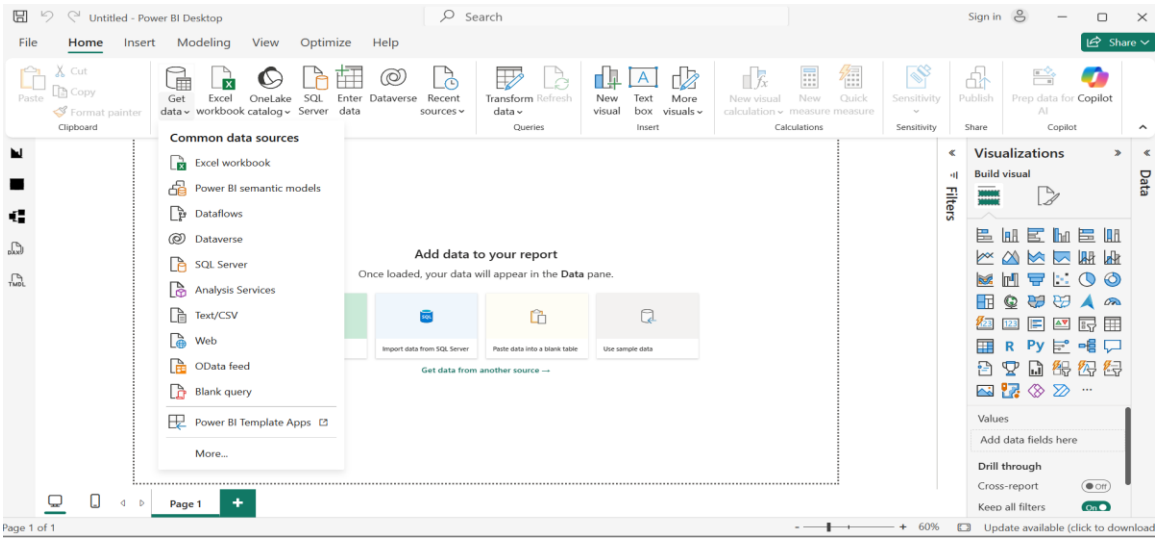
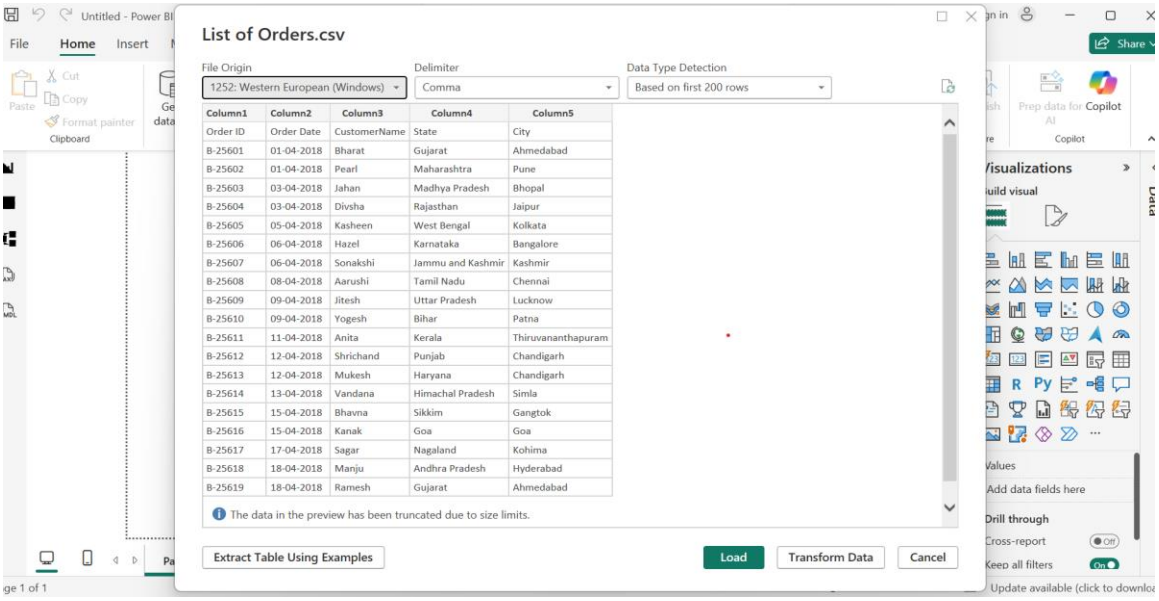
Power BI Assignment 1

E Commerce Sales Analysis

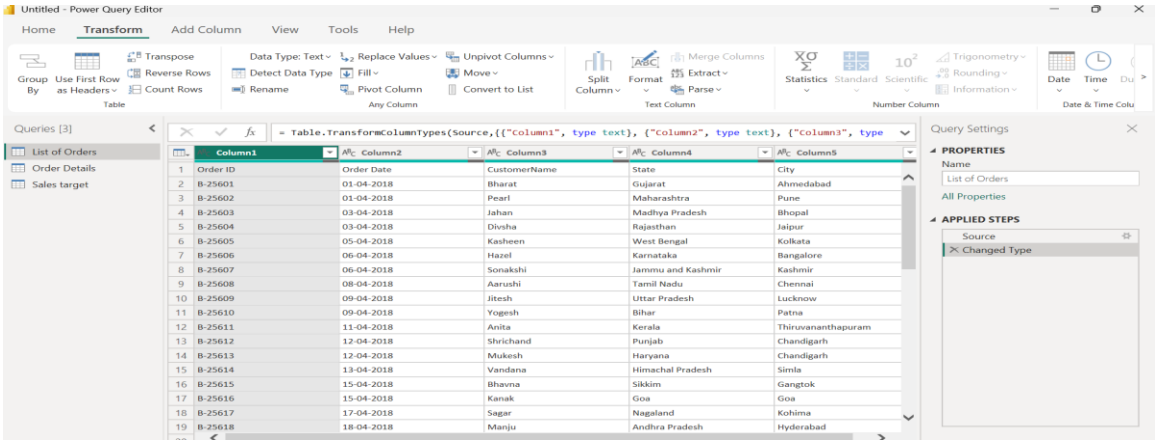
Import Data:

Import “List of Orders.csv” into Power BI.



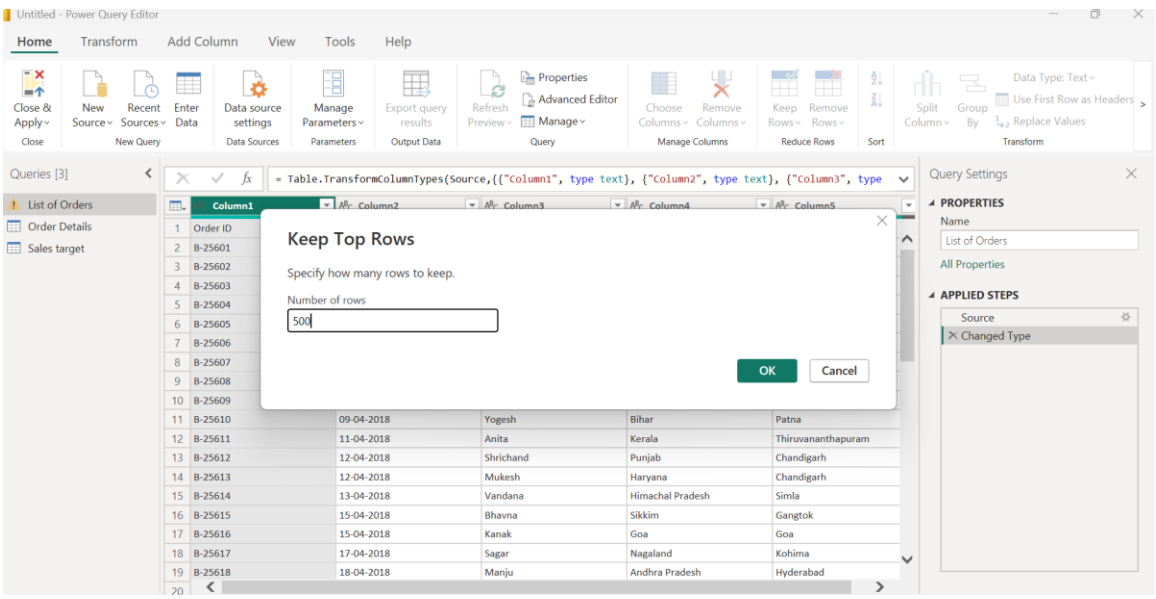


Open “List of Orders” in Power Query Editor by clicking on **‘Transform’.**

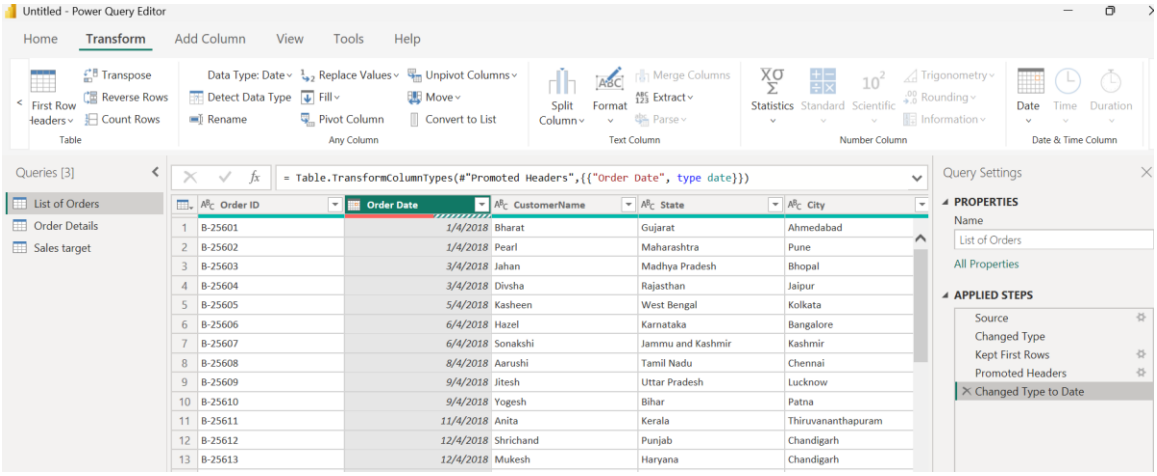


Data Transformation:

The "List of Orders" table to only the **first 500 rows**.

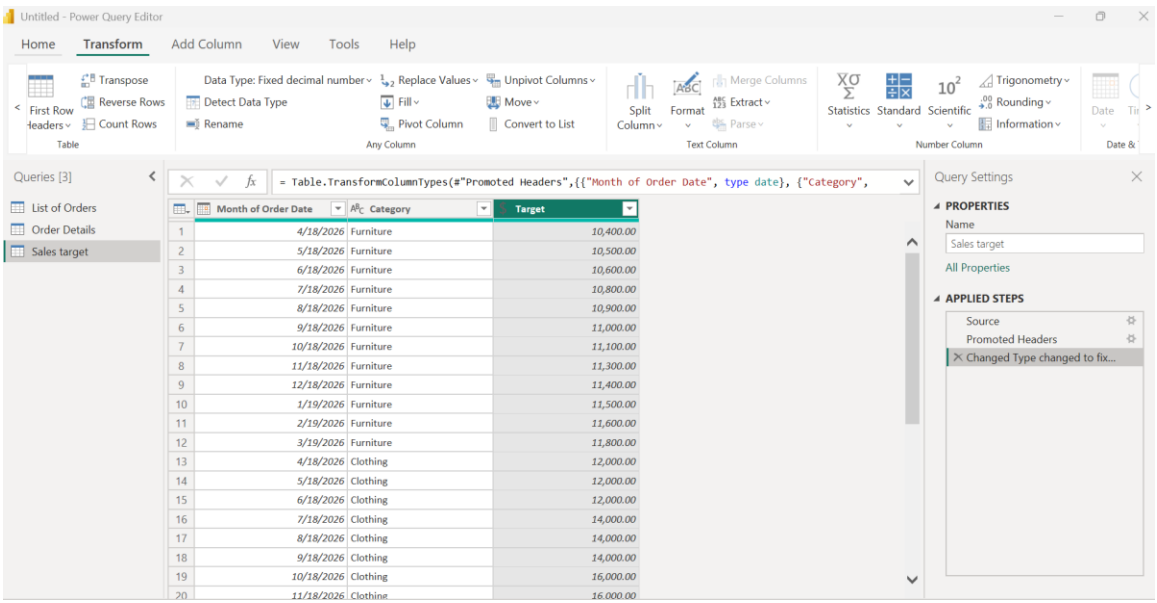


The “Order Date” column in the “List of Orders” table is set to data type **'Date'.**

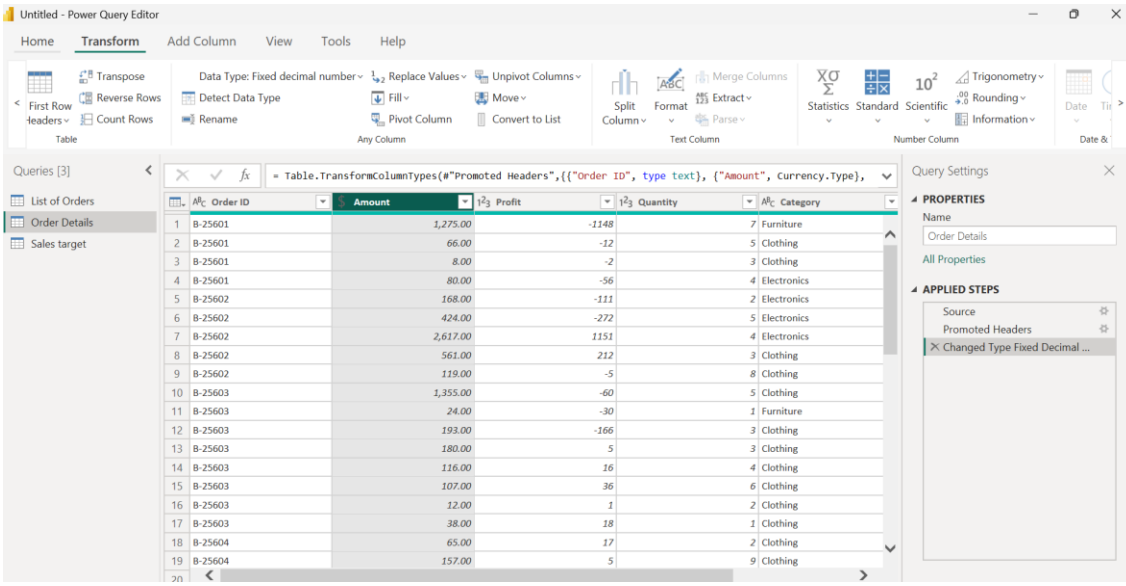


	Order ID	Order Date	CustomerName	State	City
1	B-25601	1/4/2018	Bharat	Gujarat	Ahmedabad
2	B-25602	1/4/2018	Pearl	Maharashtra	Pune
3	B-25603	3/4/2018	Jahan	Madhya Pradesh	Bhopal
4	B-25604	3/4/2018	Divsha	Rajasthan	Jaipur
5	B-25605	5/4/2018	Kasheen	West Bengal	Kolkata
6	B-25606	6/4/2018	Hazel	Karnataka	Bangalore
7	B-25607	6/4/2018	Sonakshi	Jammu and Kashmir	Kashmir
8	B-25608	8/4/2018	Aarushi	Tamil Nadu	Chennai
9	B-25609	9/4/2018	Jitesh	Uttar Pradesh	Lucknow
10	B-25610	9/4/2018	Yogesh	Bihar	Patna
11	B-25611	11/4/2018	Anita	Kerala	Thiruvananthapuram
12	B-25612	12/4/2018	Shrichand	Punjab	Chandigarh
13	B-25613	12/4/2018	Mukesh	Haryana	Chandigarh

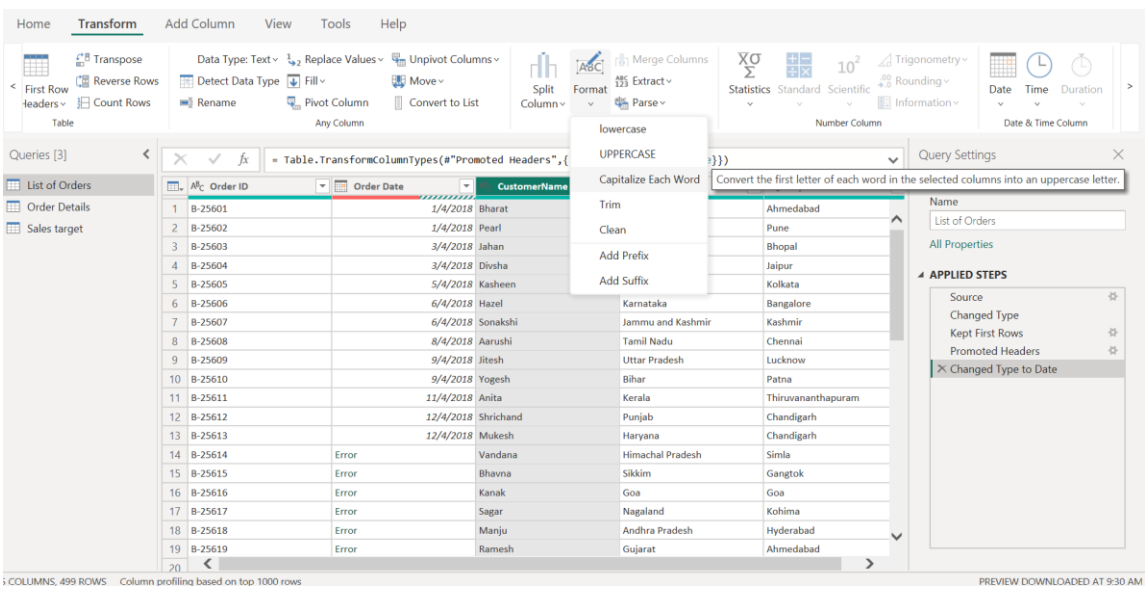
Change the data type of “Amount” and “Target” columns to **‘Fixed Decimal Number’**.



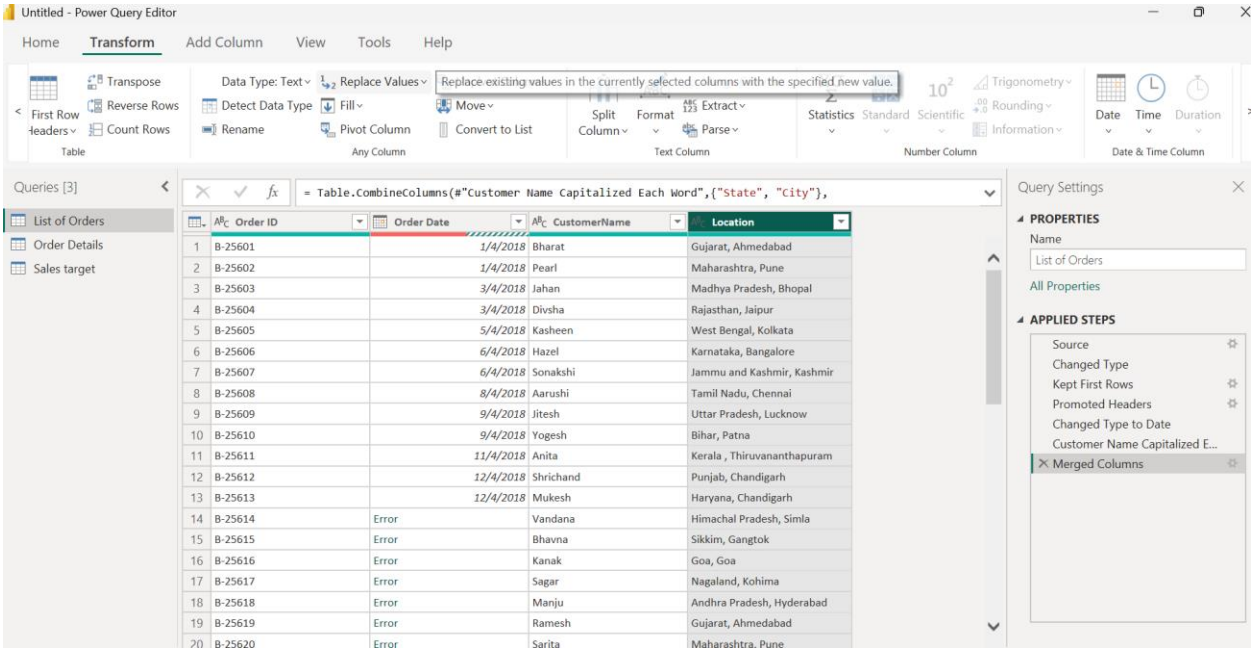
	Month of Order Date	Category	Target
1	4/18/2026	Furniture	10,400.00
2	5/18/2026	Furniture	10,500.00
3	6/18/2026	Furniture	10,600.00
4	7/18/2026	Furniture	10,800.00
5	8/18/2026	Furniture	10,900.00
6	9/18/2026	Furniture	11,000.00
7	10/18/2026	Furniture	11,100.00
8	11/18/2026	Furniture	11,300.00
9	12/18/2026	Furniture	11,400.00
10	1/19/2026	Furniture	11,500.00
11	2/19/2026	Furniture	11,600.00
12	3/19/2026	Furniture	11,800.00
13	4/18/2026	Clothing	12,000.00
14	5/18/2026	Clothing	12,000.00
15	6/18/2026	Clothing	12,000.00
16	7/18/2026	Clothing	14,000.00
17	8/18/2026	Clothing	14,000.00
18	9/18/2026	Clothing	14,000.00
19	10/18/2026	Clothing	16,000.00
20	11/18/2026	Clothing	16,000.00



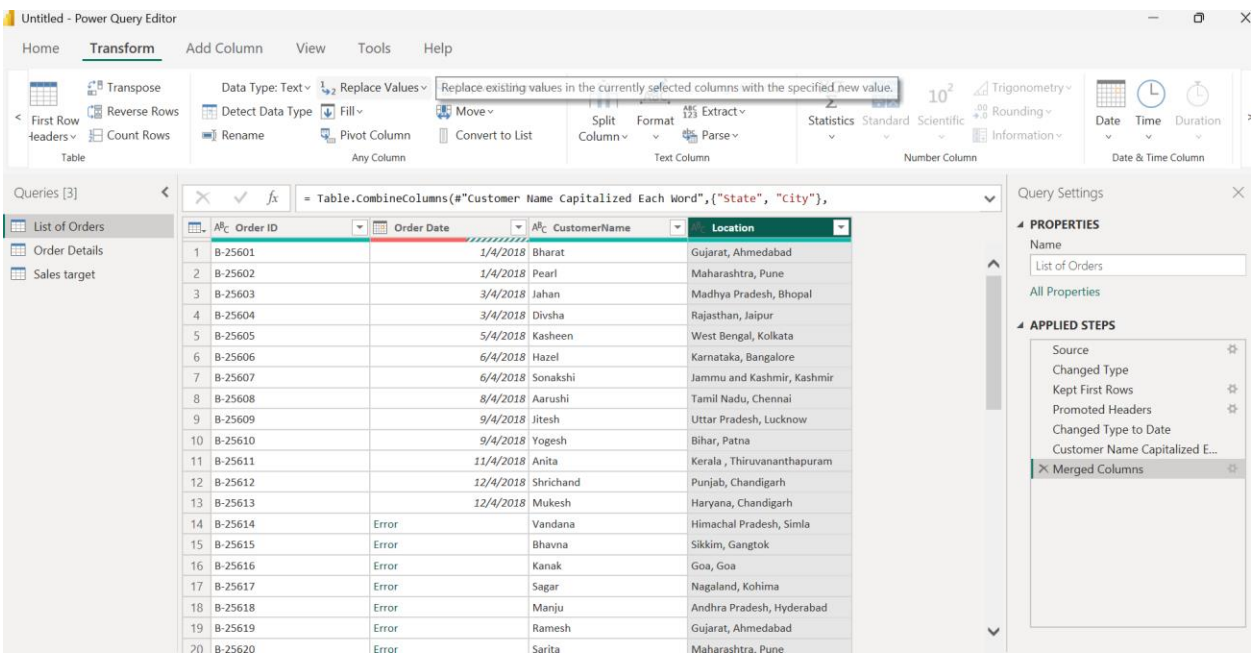
Format the **"Customer Name"** column into proper case, ensuring consistent **capitalization for each word.**

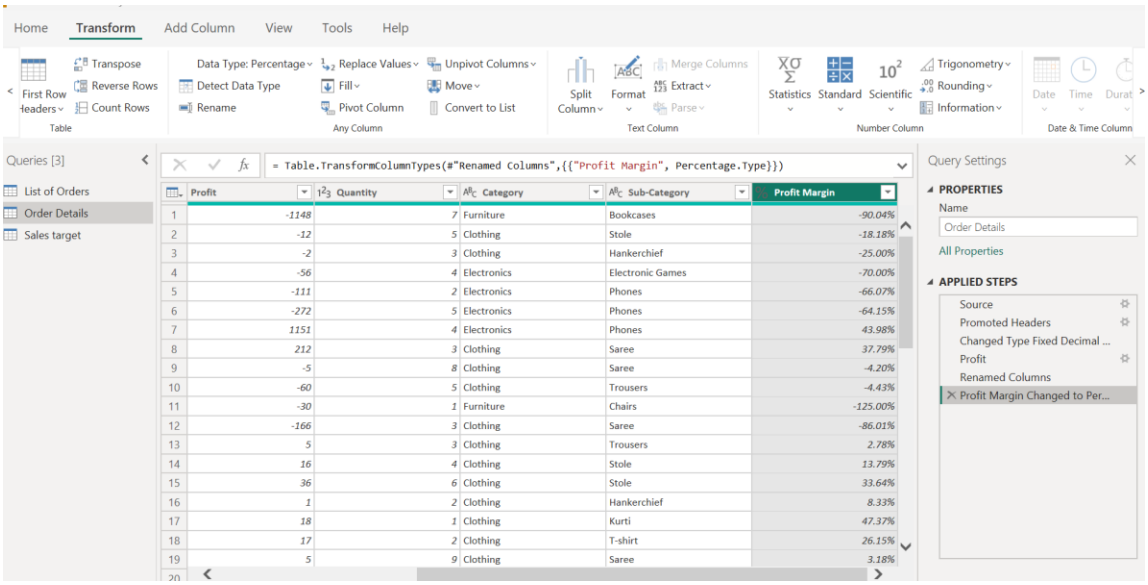


Merge the "State" and "City" columns to create a new column named **"Location"** in the format ‘City, State’.

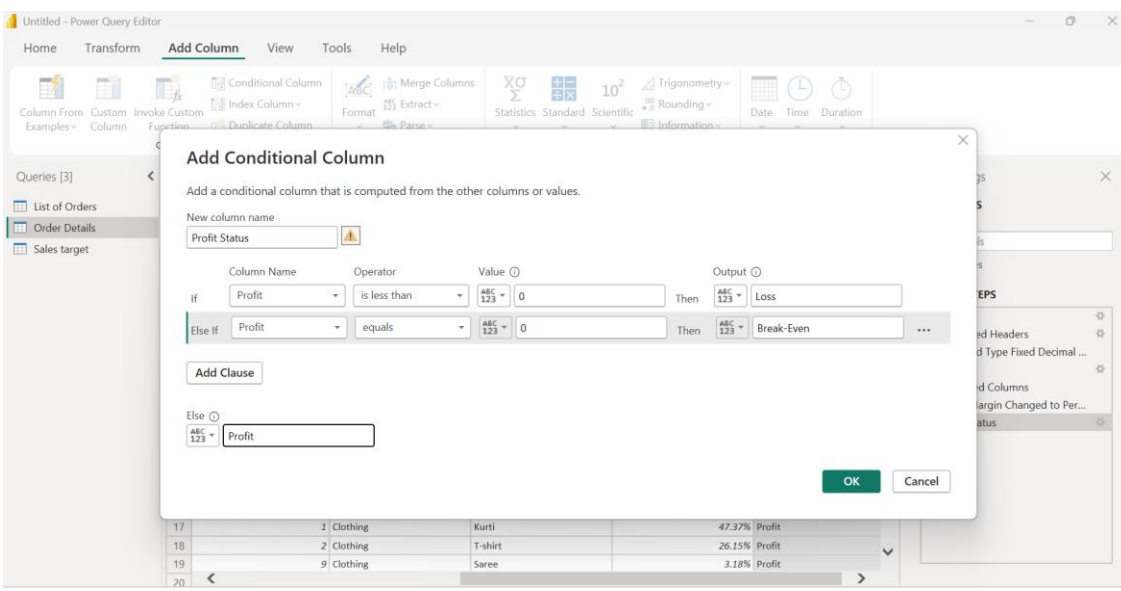


Create a new custom column named **"Profit Margin"** as the percentage of "Profit" divided by **"Amount".**



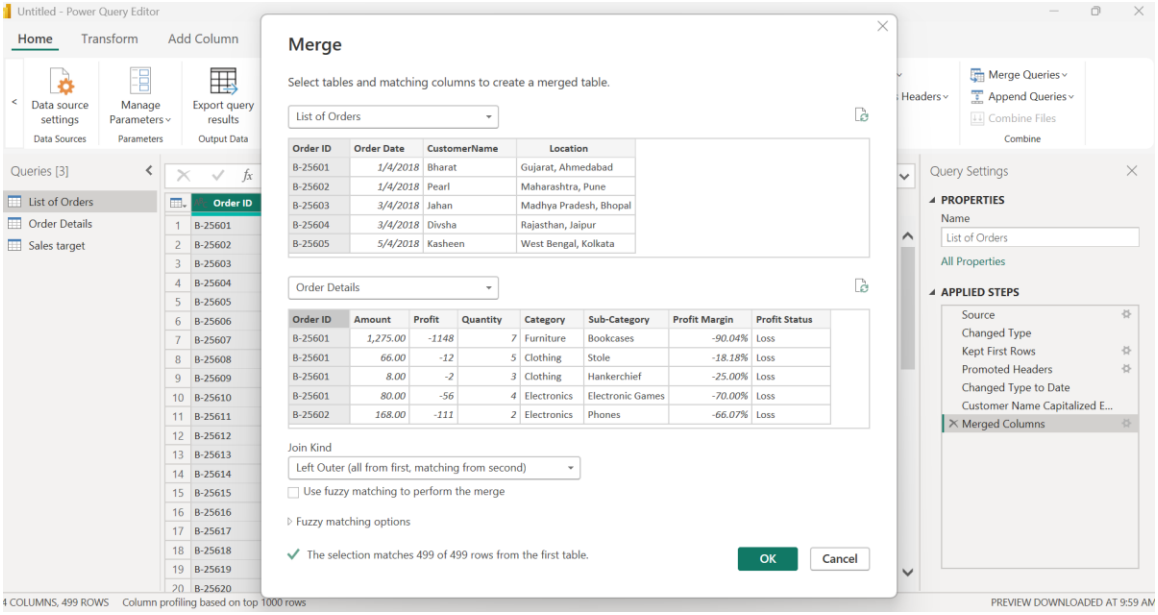


Add a new conditional column named **"Profit Status"** based on the values in the "Profit" column. The conditions are as follows: if the profit is less than 0, the label should be "**Loss";** if the profit equals 0, the label should be **"Break-Even";** and if the profit is greater than 0, the label should be **"Profit".**



Merging Data (Joins):

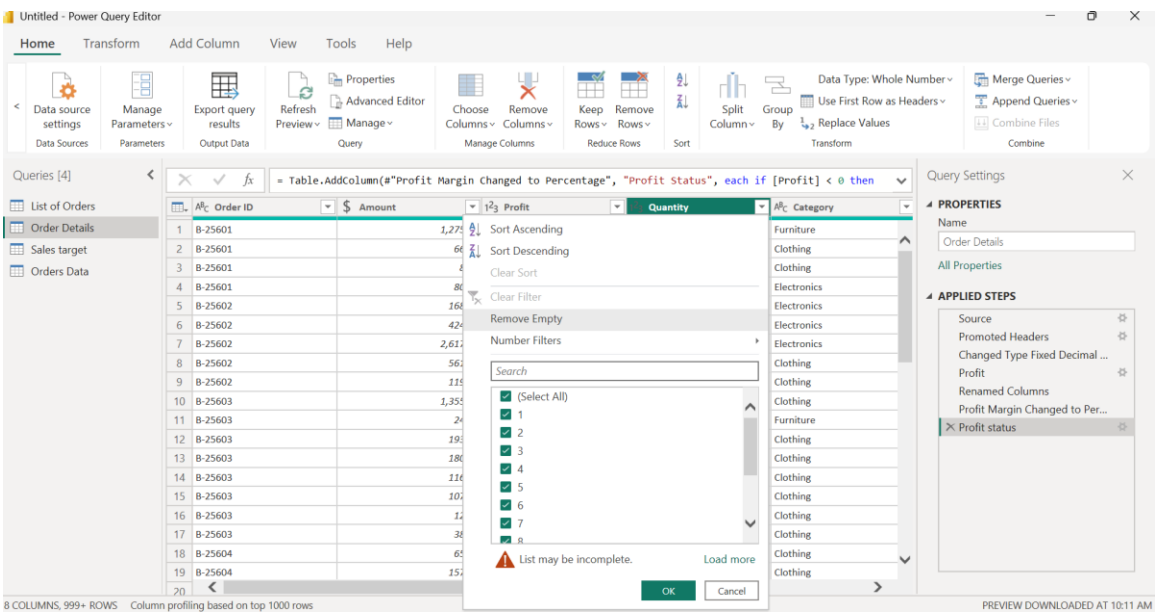
Merge the "List of Orders" and "Order Details" tables into a new single table named "**Orders Data"** based on the "Order ID" relationship.

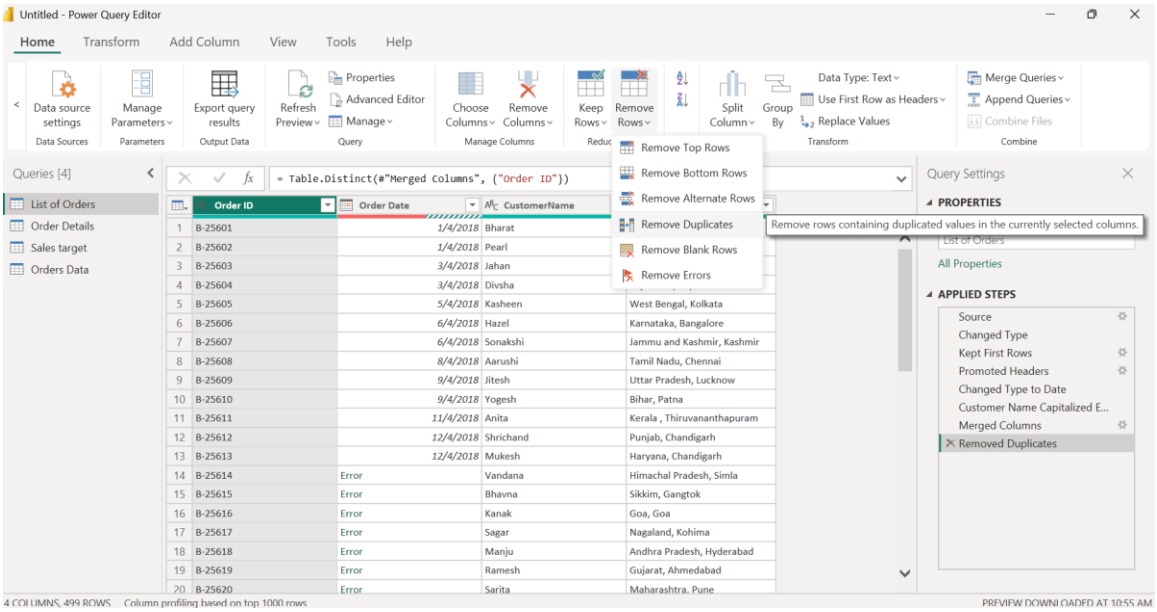


Handling Missing Data & Duplicate Data:

Identify **missing values** in the data and determine a strategy to address them.

Check for duplicate rows and define a strategy to handle **duplicates.**



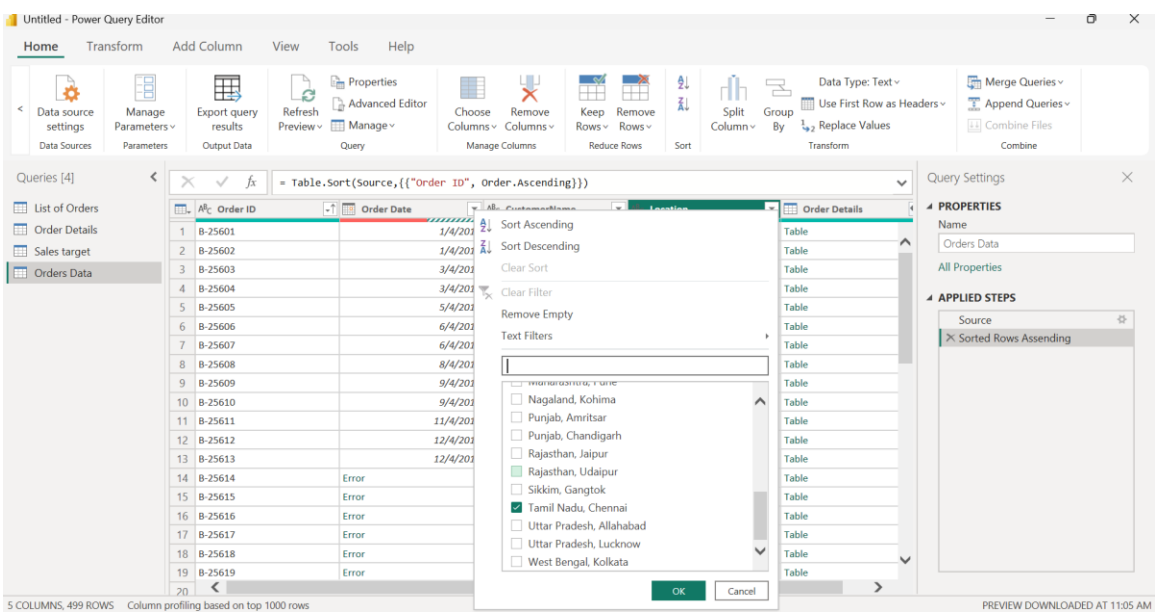


Sorting and Filtering Data:

● In the ‘Orders Data’ table, utilize sorting and filtering techniques on columns like Order Date, State or Category to analyze data based on specific criteria:

 Sort the orders by Order Date in descending order to analyze recent trends.

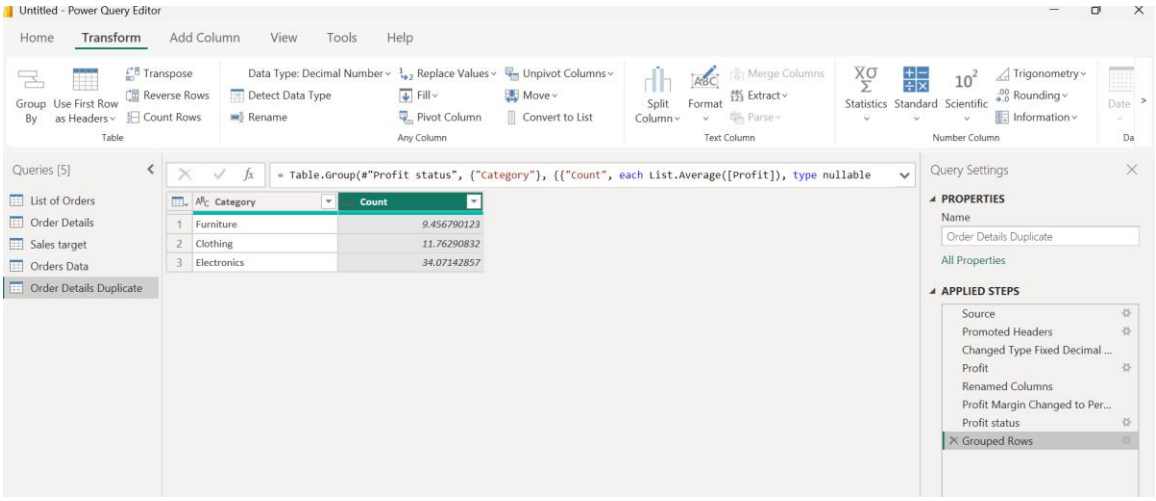
 Filter the orders to focus only on a specific state (e.g., Tamil Nadu) for regional analysis.

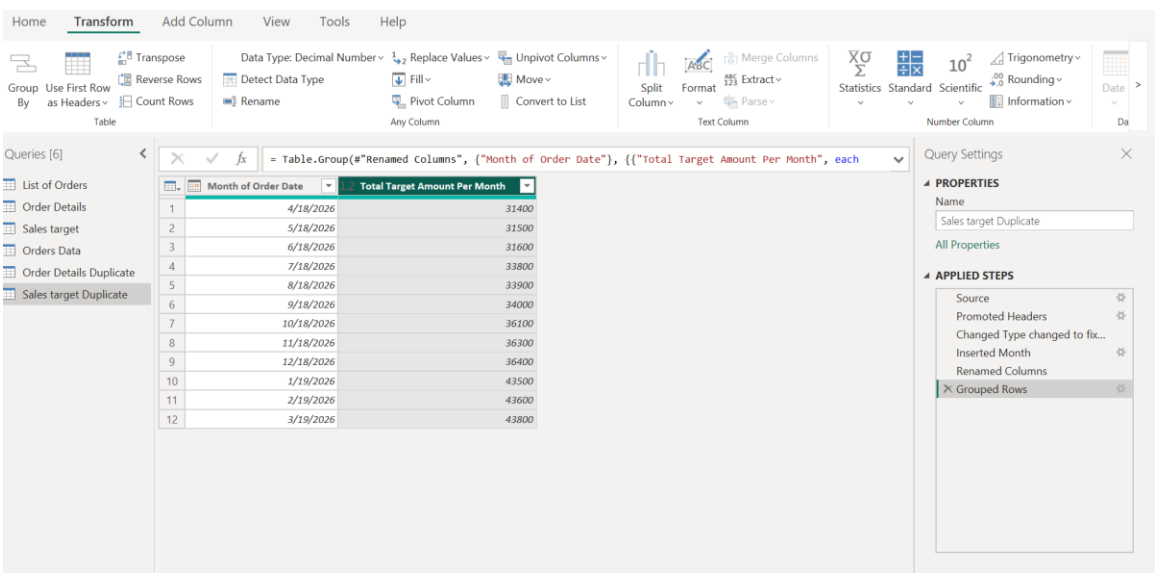


Grouping and Aggregating Data:

Duplicate the “Order Details” table and calculate the count of each Order ID, average profit by Category or total amount by Sub-Category.

Duplicate the “Sales Target” table and aggregate the total target amount by Month of Order Date.

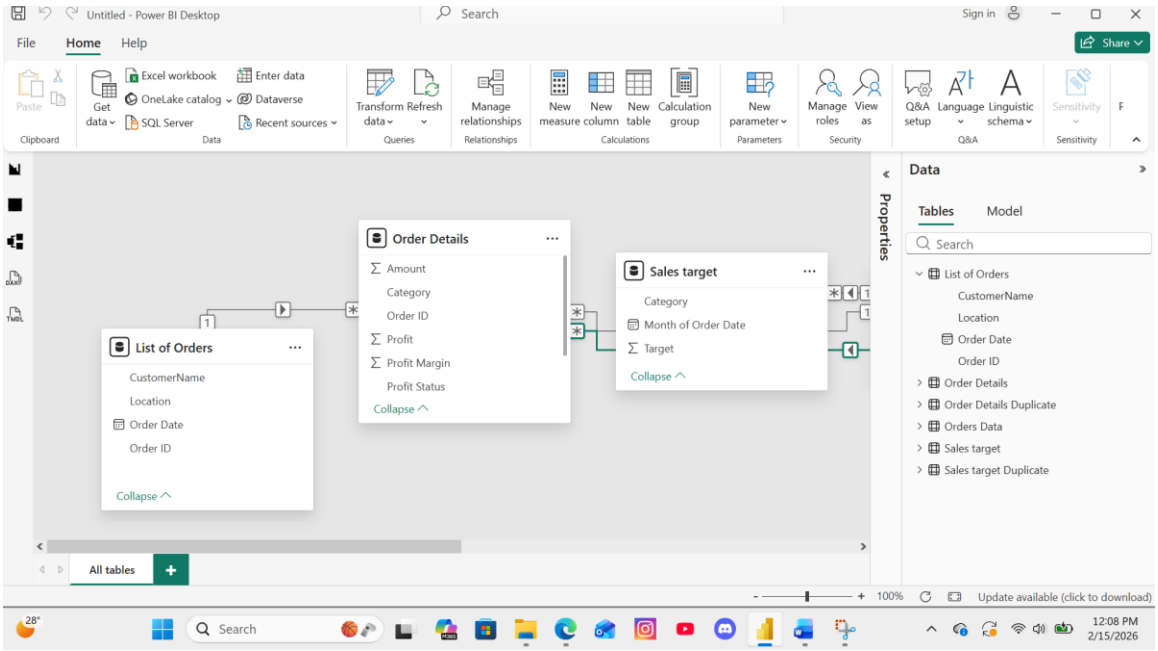


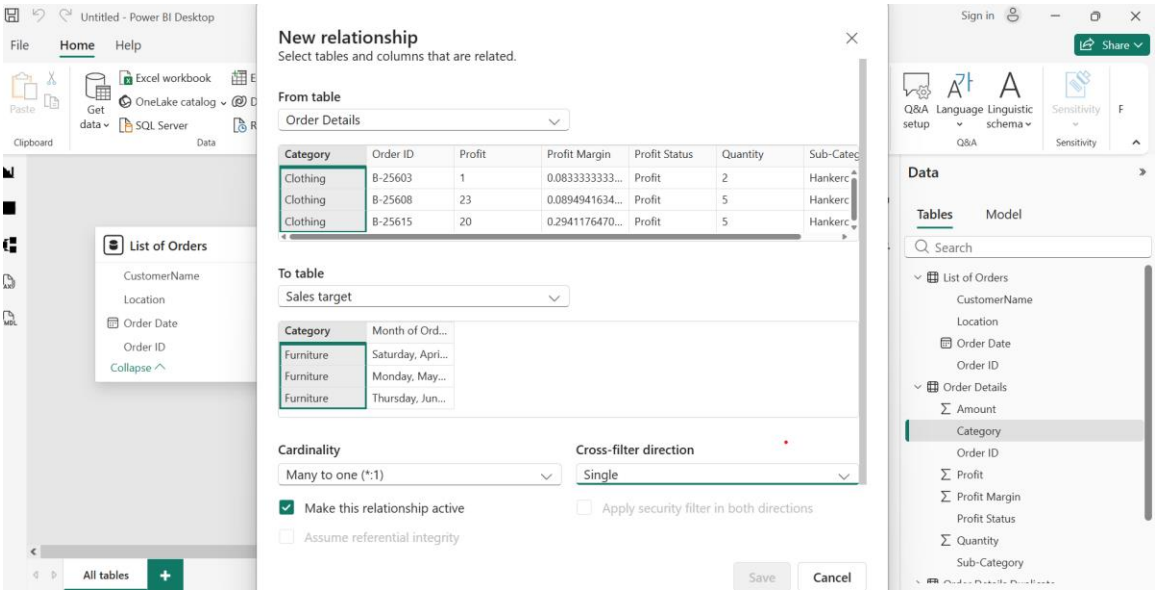


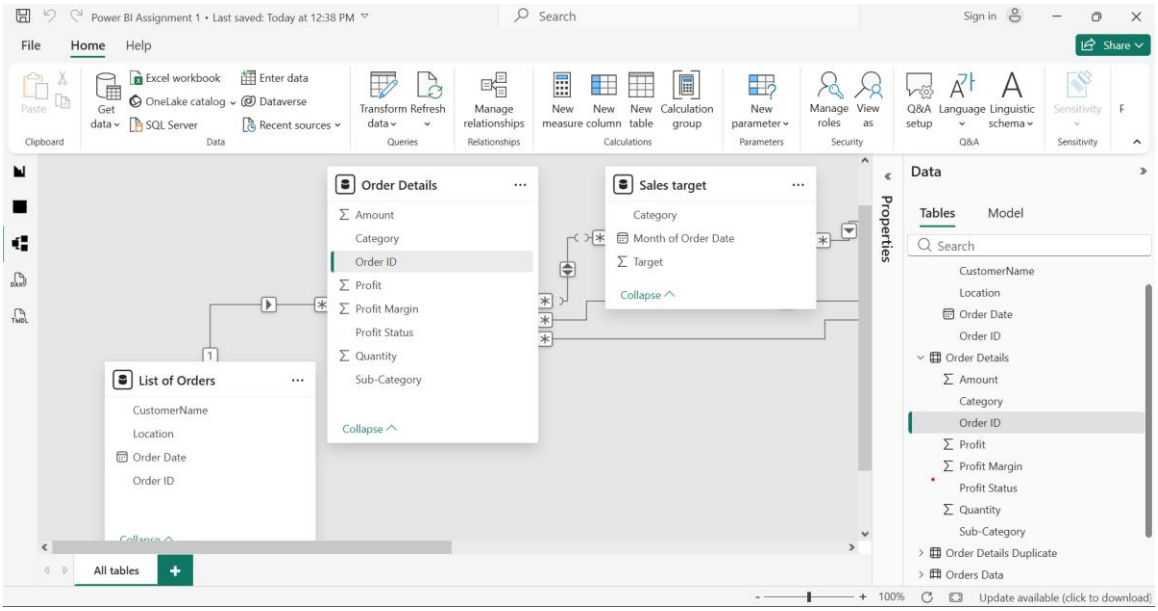
Data Modeling:

Establish a relationship between the “List of Orders” and “Order Details” tables using the ‘Order ID’ column.

Build a relationship between the “Order Details” and “Sales Target” tables based on the ‘Category’ column. Click "Manage relationships" and ensure this relationship is active.






Power BI Assignment 1 • Last saved: Today at 12:38 PM
Search
Sign in
File
Home
Help
Share
Paste
Clipboard
Get data
Excel workbook
OneLake catalog
SQL Server
Enter data
Dataverse
Recent sources
Data
Transform data
Refresh
Queries
Manage relationships
Relationships
New measure
New column
New table
Calculation group
Calculations
New parameter
Parameters
Manage roles
View as
Security
Q&A setup
Language
Linguistic schema
Q&A
Sensitivity
Order Details
Amount
Category
Order ID
Profit
Profit Margin
Profit Status
Quantity
Sub-Category
Collapse
Sales target
Category
Month of Order Date
Target
Collapse
List of Orders
CustomerName
Location
Order Date
Order ID
Properties
Data
Tables
Model
Search
CustomerName
Location
Order Date
Order ID
Order Details
Amount
Category
Order ID
Profit
Profit Margin
Profit Status
Quantity
Sub-Category
Order Details Duplicate
Orders Data
All tables
100%
Update available (click to download)